Measuring Boards Using Quantitative Tools from Natural Language Processing

Lewis Hooper

Abstract

Natural language processing (NLP) tools provide quantitative methods to analyze board minutes and better understand and measure the work of the board. Techniques such as riverbed graphs and sentiment analysis provide objective, measurable information about key areas (finance, quality, compliance) that boards focus their attention on. By comparing the key focus areas between different hospitals, this paper demonstrates that NLP tools can provide a robust and reproducible way in which boards can measure their progress toward the betterment of their goals, objectives and responsibilities for both peer-to-peer and internal comparisons.

Introduction

Hospitals and healthcare are highly focused on measuring the quality and performance of their services. This focus on measurement started in clinical care but now extends to almost all parts of the health system. Over time, the measurement tools used have evolved to include increasingly quantitative measures of both process and outcome. The one area where there are limited quantitative measures is at the board level (Mays et al. 2007; Peck 1995).

There is a need to develop board performance guidelines (Auditor General of Ontario 2008) that look at board function and compare that performance against other boards in a quantitative way. Natural language processing (NLP) tools provide useful analytic methods to review board minutes and provide insight into the board's focus. The database developed for this analysis contains over 1,200 meetings, including over 6,000 motions from 53 hospitals across the jurisdictions of Ontario, Scotland and New Zealand.

This paper focuses on Ontario hospitals as a means of introducing the topic and showing the potential usefulness in understanding which areas the boards focus on and the development of benchmarks for them. The Ontario data were drawn from the websites of 22 acute hospitals, including a mix of academic, urban and rural hospitals. The Ontario data include over 700 meetings and 3,000 motions, averaging 33.9 meetings per hospital and ranging from a year to 10 years' worth of data. Information from boards in Scotland and New Zealand is used to highlight differences but is not included in the analytic sections.

Boards have a number of key roles in an organization (Corbet and Pessione 2019; Quigley and Scott 2004). These include the general fiduciary responsibilities; responsibility for organizational leadership and oversight; ensuring compliance with mandated regulations; the development of and/or approval of foundational documents; mission, vision and values; and the development and monitoring of strategic plans. Ontario boards are also responsible for board policies that may guide the board and the organization, recruitment of board members and the appointment and oversight of medical staff. There are additional episodic foci for boards, such as the oversight of special fundraising projects, major capital projects and other special projects. This paper focuses only on the baseline issues common across all boards, such as finance, quality, mission, vision, values, policy and bylaws, strategy and compliance, and the common tools used by boards (in-camera sessions, consent agendas and executive committees).

There are little or no objective performance criteria regarding what a board focuses on...

The process used by boards with respect to these foci is similar across all organizations. Boards are deliberative bodies that: (1) discuss and review issues of concern, including those identified by the board and those brought to the board by management, and (2) provide direction to the organization and the board itself. Direction includes both formal motions of the board and less formal suggestions to management. These two actions, namely, deliberation and directions, are recorded in board minutes. There is considerable variation in the organization and detailing of the minutes. However, minutes almost always describe the issue, the background, relevant discussion and any action the board takes concerning that issue. In Ontario, when the board directs or approves a specific action, it is almost always done with a specific motion.

There are little or no objective performance criteria regarding what a board focuses on or how it balances those foci. There are self-assessment tools, and many of the boards in the study have used those tools. These tools are useful but are perhaps too dependent on the opinion of board members (Peck 1995). Accreditation Canada provides an excellent self-assessment system followed up with an external review, which includes governance, based on generally accepted principles of best practice that outline suggested activities and mechanisms that should be in place (https://accreditation.ca/accreditation/qmentum/).

However, there are no known tools to measure hospital board performance in a data-driven manner using the board's own words to document its activities. The literature documents the use of NLP tools to analyze the minutes of central banks and other financial institutions (Apel and Grimaldi 2012) and those of the US Food and Drug Administration (Shen et al. 2018), but the use of these tools in other areas has not been reported.

General Methods

Board minutes were downloaded from the websites of selected hospitals, 22 in Ontario, 18 in New Zealand and 13 in Scotland. The minutes were converted to machine-readable text using open-source tools. In many cases, the minutes were extracted from larger documents that included agendas and background information not useful in this analysis. Multiple instances of the minutes were created, including a full copy for reference and other copies processed in various ways depending on the required analysis. The processing included lemmatization, removal of stop words, various tokenization methodologies and the latent Dirichlet allocation (LDA), which facilitated the selection of board foci and terms associated with those foci. Sentiment analysis was done through the application of the National Research Council Word-Emotion Association Lexicon (NRC EmoLex) tool (NRC 2016). Hospital and board member names were removed to facilitate frequency analysis using the Name data set (Remy 2020). The entire code was written in R, and all the software used is open source.

Observations and analysis (Ontario specific)

The first use of the data is the development of a riverbed graphic determining the relative focus on key topics for the 22 Ontario hospitals as a whole, then comparing that with the focus of individual hospitals. The foci and associated terms were identified through LDA topic analysis and literature reviews (Auditor General of Ontario 2008; Barnett and Clayden 2007; Corbett and Mackay 2013; Corbett and Pessione 2019; Ditzel et al. 2006; Quigley and Scott 2004, 2005). The basic graphic shown in Figure 1 contains the summary for the 22 Ontario hospitals based on 747 board meetings.

The thick black line in the graph shows the mean percentage of focus by item for the 22 hospitals, and the shaded area, representing the banks of the river, shows the variation around that mean. The top of the shaded area represents the 80th percentile for that focus group, while the bottom edge represents the 20th percentile. The foci are arranged in ascending order, with the least frequently referred to foci on the left-hand side and the most frequently discussed foci on the right-hand side, showing that, on average, finance constitutes about 17% of the board focus. By contrast, quality constitutes about 4% of the board focus.

The high number for finance is consistent with those for Scotland and New Zealand, where the other two jurisdictions have a slightly higher focus than Ontario.

The numbers for the management-related focus are highly variable. In some boards, the management team is rarely referred to, and in others, management seems to play a key role in the deliberations of the board. In this area, Ontario is higher than the other jurisdictions.

Some items, although important, are dealt with infrequently and thus are a lower proportion of the overall focus (mission, vision and values). Others (quality) represent the tip of an iceberg of an ongoing effort that comes to the board only

FIGURE 1.
Percentage of board focus by area

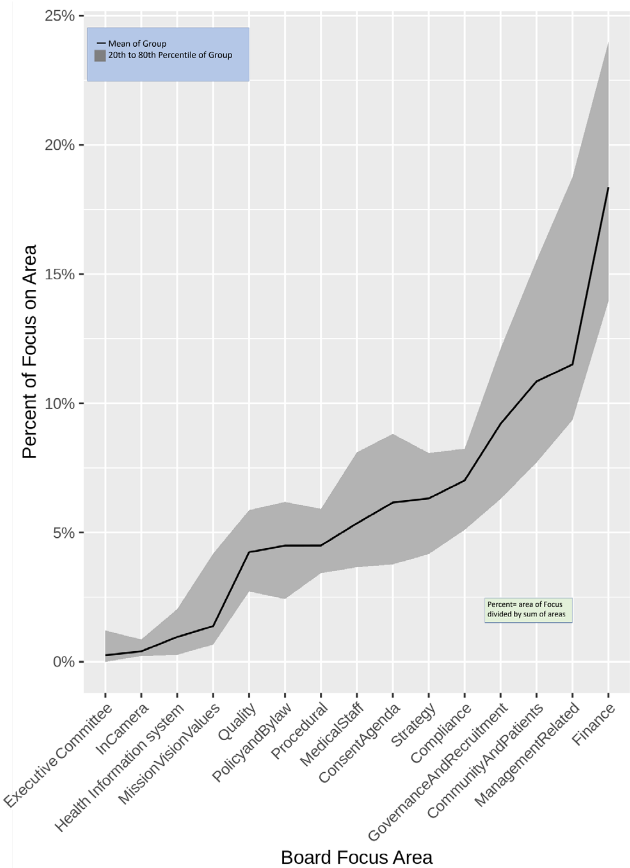
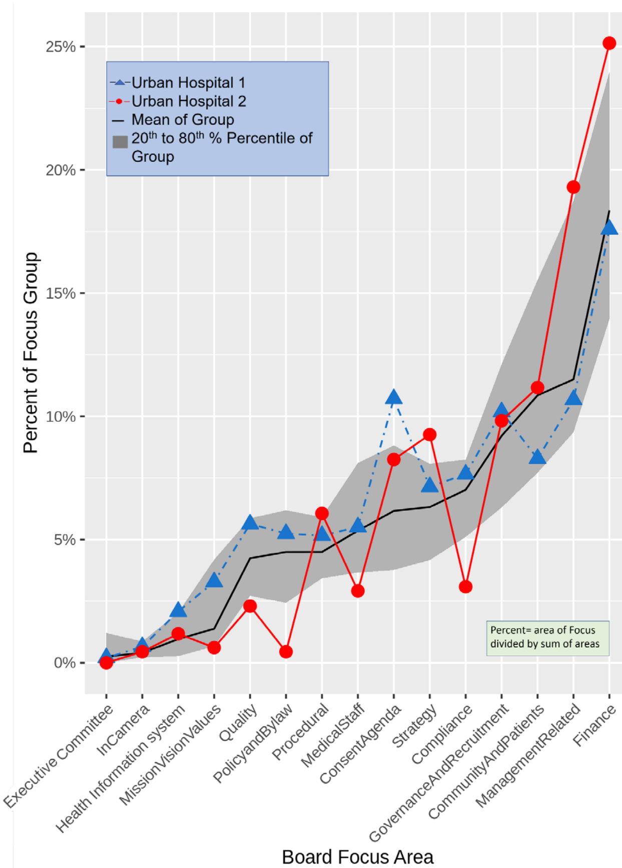


FIGURE 2.
Comparing two urban hospitals with the Ontario average



for final review.

In general, the riverbed graphic represents the norms for the Ontario hospitals. The order and the variation around the foci are generally consistent with what happens in other jurisdictions.

As shown in Figure 2, an individual hospital's focus areas can be compared to those of the group. This allows the hospital to gauge how their board differs from that of the group.

The two hospitals overlaid on the riverbed graphic in Figure 2 provide illustrative examples of how a board can analyze its work and better understand where it may wish to focus its limited time.

Urban hospital 1 is generally within the variation used in this paper but does have a high use of consent agendas. Other external data points, including financial performance and Canadian Institute for Health Information (CIHI) hospital standardized mortality rate (HSMR), show urban hospital 1 to be performing about average.

Urban hospital 2 is much more variable and shows several areas outside the banks. There is a high focus on finance, management and strategy. There is a low focus on mission, vision and values; quality; policy and bylaw; medical staff; and compliance. External data show this hospital to be performing well with respect to finance but performing poorer than the average with respect to HSMR.

Urban hospital 1 may want to review its use of the consent agenda, but, overall, it seems to be performing in a manner quite similar to the rest of Ontario hospitals.

The board in urban hospital 2 may wish to explore the effort it puts into compliance, that is, is it meeting the legal obligations and responsibilities of the board? If so, then perhaps the mechanisms used could be shared with those hospitals that have a greater focus in this area. The low focus on policy and bylaws and the high focus on management may indicate that the balance between the board and management should be explored. Given its positive financial performance and poorer performance in HSMR, the board in urban hospital 2 may wish to refocus some of its financial energy on quality.

Each individual board could use this information as part of its ongoing self-analysis to see if there are opportunities to improve its contribution to, and oversight of, the organization. Linking this work to the sentiment analysis (next section) may be helpful.

In addition, for boards that decide to change their focus, analyzing the riverbed graph over time would show the board how effective it has been in implementing the changes.

Sentiment analysis

A second use for the data is sentiment analysis, which may illuminate board dynamics. By itself, sentiment analysis should not be taken as a performance measure, and there are limitations on its use (explained later). However, it has value on its own and – when coupled with the riverbed graph – may help the board analyze and improve its performance. Originally, sentiment analysis was used when analyzing online comments to quantitatively assess those comments (Pang and Lee 2002). Sentiment analysis has been extended to other documents (Behdenna et al. 2018), and the National Research Council (NRC 2016) of Canada has developed tools to analyze the sentiment of documents. The NRC EmoLex tool may be particularly useful in analyzing minutes (NRC 2016). The tool outlines eight emotions and two sentiments. For this review, the analysis was limited to the emotion of trust and the two sentiments – positive and negative.

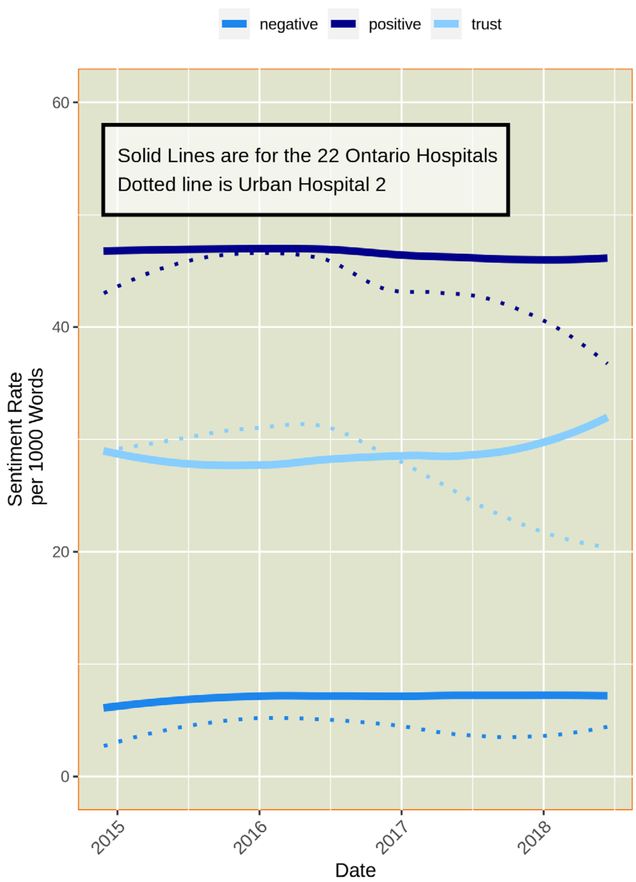
The NRC EmoLex tool defines a list of words associated with one or more of the emotions or sentiments. The EmoLex list was modified slightly for use in this review to better reflect the health industry. For example, in the general public, the word "hospital" carries an emotion of trust, while "discharge" carries a negative sentiment. Those words and other similar words carry a different context in the healthcare domain and were excluded from the review. Frequency analysis for each topic was determined for every meeting. The data are presented in Figure 3 (sentiment analysis) and show a relatively stable mean pattern of trust and positive and negative sentiments over time for the Ontario hospitals. Figure 3 also shows the sentiment analysis for urban hospital 2, which is reviewed in the section on riverbed graphics. Urban hospital 1 is close to the pattern of all the Ontario hospitals and is not shown. All sentiment data have been smoothed, and the data for Ontario were trimmed to match the time-specific data from urban hospital 2.

Urban hospital 2 shows a continuing decline in trust and positive sentiment beginning in 2016. There are several potential reasons for this variation, but it should give the board a point of reflection. When combined with the riverbed graphic, there are issues that the board may wish to consider. Is the board focused on the right things? Is there dissatisfaction with the board's process, environmental issues, etc., or is this simply a matter of a change in recording style?

Summary Application and Implementation

The riverbed graphic could be used by a board and senior management as a yearly benchmark for where the board focused and how that focus compares with the other boards. This could be part of the board's annual review and would serve to help the board understand if it is focusing on the relevant parameters for that board. As noted elsewhere, this could be done in conjunction with other external performance markers, such as CIHI or Health Quality Ontario quality markers, or financial benchmarks and the board's judgement on where it should focus. The riverbed graphic is important on a more frequent basis if the board wishes to change its focus. For example, if a board decides to focus more on quality, the riverbed graphic could be generated on a monthly or quarterly basis and would help the board measure its progress.

FIGURE 3.
Overall sentiment at Ontario hospitals



Sentiment analysis is a tool that may provide insight into board and board-management dynamics. Once set up, it would be a useful complement to the monthly board meeting evaluation that many boards undertake. Sentiment analysis is probably best used to monitor trends over time, and a three-meeting rolling average is likely to be more useful than the score of an individual meeting. A yearly summary showing how the board's overall sentiment compares to that of the other hospitals may be useful, but only if there is a significant variation from the peer group or if there are unusual trends.

There are other potential uses for NLP tools, which are discussed in the section "Next Steps." These potential benefits could be achieved in several ways, including the expansion of the toolset beyond the riverbed graphic or sentiment analysis. In addition, the toolset could be used in other parts of the organization to understand focus, sentiment or other concepts.

Developing and implementing these tools at the individual hospital level could be problematic. In particular, the tasks related to data gathering and cleaning and preparation of the required information for benchmarking purposes should not be understated. Furthermore, the creation of the riverbed graphic and sentiment analysis from clean data requires familiarity with skills and programming languages not normally found in most hospital decision-support teams.

There are three potential options for implementing these tools:

1. Complete in-house development: This will be a labour-intensive effort, especially gathering, cleaning and preparing the minutes from other hospitals' or agencies' websites. It would give the hospital a lot of flexibility in finding new uses for the tool but is probably not practical for most Ontario hospitals.
2. Shared/outsourced implementation: A group of hospitals or a service provider could undertake the work. This should make the data collection much simpler, and if a critical mass of participants is achieved, then the analytic work could be done at a fairly low unit cost.
3. An external agency could undertake the collection and analysis (such as the Ontario Health Association, CIHI). This would drive the unit cost down but might be less innovative. NLP is just beginning to be applied to this type of work, and it might be prudent to have a flexible approach.

Hospital boards are at the apex of a complex and expensive system.

Helping boards ensure that they are doing the right things, and enabling them to fine-tune their governance approach will provide a significant benefit to the health system.

Limitations

This work is limited to publicly available minutes from a subset of Ontario hospitals. It does not include in-camera or closed-session minutes, except for those cases where those minutes have been made public. Currently, the work does not use data from board committees.

Variations in how an individual hospital uses the consent agenda may impact the overall focus represented in this analysis. A consent agenda can inadvertently hide decisions made by the board. For example, a consent agenda that "receives and approves the recommendations in the committee minutes" and does not publish that discussion will mean that those foci are unavailable for benchmarking.

Ontario hospitals sometimes eventually release decisions made in-camera, but most do not, and, in fact, the topic of discussion is usually hidden. In contrast, New Zealand uses a highly structured approach for what it terms publicly excluded sessions. This process protects the board's deliberations and decisions but does identify the general focus of those discussions.

Some tools used by boards, executive committees, consent agendas and in camera sessions, may enable a board to run more efficiently. However, in Ontario, a lack of consistency in their usage may be problematic when benchmarking boards.

The sentiment analysis tool, as used, is a general tool, and testing and further refinement of the NRC EmoLex tool for use in hospitals will improve the tool. The variability in the recording style for minutes likely has more impact on sentiment analysis than on the riverbed graphic. Some hospitals record minutes in a sparse manner, focusing on decisions, deadlines and dollars, which may influence sentiment analysis.

The recorder and the initial reviewer may modify or mute the originally spoken words.

This risk is mitigated by the requitement that minutes are approved by the full board, and errors or omissions can be corrected by board members.

Conclusion

The two methods presented here are examples of the ways NLP tools, including the riverbed graphic and sentiment analysis, could help boards understand their performance and compare it with a benchmark. These tools show where a board is focusing its attention and the overall tone of the board over time. This work should be taken as a preliminary guide but represents an important step in improving the performance of boards.

Each board will set its plans and agendas based on its view of the mission and issues faced by the board. The riverbed graphic provides the board with a view of where it is at variance with other boards. When a board is outside the banks of the variance, that board can decide if it wishes to rebalance its workload. The riverbed graphic can also be used by that hospital to measure its progress toward rebalancing. For example, urban hospital 2 may have an opportunity and desire to increase its attention to the quality and policy areas, and progress on this could be measured using the riverbed graph. Other hospitals will have different opportunities and will make different decisions.

Sentiment analysis may be useful as an overall measure of the cohesiveness of the team (board, management and medical staff). Monitoring the trends in sentiment over time can show potential issues. A trend showing a decline in trust and positive sentiment may show the need for an internal review.

There are some instances in the data where declining trust and positive sentiment precede an unexpected chief executive officer (CEO) turnover. This observation is one of the areas where further analysis should be done and is outlined in the "Next Steps" section.

Neither the riverbed graphic nor sentiment analysis should be expected to identify issues on a month-to-month basis. Still, over time, both could be extremely valuable to the board.

Next Steps

There are several areas where NLP tools can be further developed. A deeper analysis of the motions of the board should be done, perhaps classifying the motions into a common, system-wide typography and allowing a deeper comparison of how boards differ in terms of emphasis.

Including the corporate statement of mission, vision and values into an analysis might be helpful. A preliminary review mapping the language of those fundamental documents onto the minutes of the board showed significant variation, with some hospitals reflecting the language of their mission, vision and values in their minutes to a greater degree than others.

Mapping the language used in the strategic plan of the hospital onto the minutes of the board may show board buy-in of the plan. The degree of board buy-in and focus may correlate with the actual implementation success of strategic plans.

There are a number of other areas where NLP techniques should be analyzed to determine if there is additional value:

- Linking the work done by board committees may show how committees impact the overall performance in the focus areas.
- The overall alignment between the board's approved mission, vision and values and strategy with the hospital as a whole could be measured by analyzing the language used in other corporate documents, news releases, memos to staff, policies and procedures, etc. This would indicate the real corporate uptake of these documents.
- A comparison with New Zealand, which utilizes specific criteria for what it calls "publicly excluded meetings," may be helpful in understanding and standardizing the use of in-camera sessions.
- An analysis to determine if the changes in trust and positive sentiment are indicators of potential CEO turnover should be undertaken.

Preliminary work in some of these areas is under way, particularly using artificial intelligence tools to classify motions. HQ

References

Apel, M. and M.B. Grimaldi. 2012. The Information Content of Central Bank Minutes. Retrieved February 1, 2020. Working Paper Series 261, Sveriges Riksbank. doi:10.2139/ssrn.2092575.

Auditor General of Ontario. 2008. Ministry of Health and Long-Term Care: Hospital Board Governance. Retrieved June 4, 2020. <https://www.auditor.on.ca/en/content/annualreports/arreports/en08/311en08.pdf>.

Barnett, P. and C. Clayden. 2007, August. *Report No. 2: Governance in District Health Boards*. Retrieved April 1, 2020. <https://www.wgtn.ac.nz/health/centres/health-services-research-centre/docs/reports/downloads/Report-No.-2-Governance.pdf>.

Behdenna, S., F. Barigou and G. Belalem. 2018. Document Level Sentiment Analysis: A Survey. *EAI Endorsed Transactions on Context-Aware Systems and Applications* 4(13): 154339. doi:10.4108/eai.14-3-2018.154339.

Corbett, A. and J.M. Mackay. 2013, September. *Guide to Good Governance: Not-for-Profit and Charitable Organizations* (2nd ed.). Ontario Hospital Association. Retrieved June 6, 2020. <https://www.gbachc.ca/BoD/orientation_manual/2.1%20-%20GuidetoGoodGovernance.pdf>.

Corbett, A. and H. Pessione. 2019, August 23. Canada: Governance Best Practices for High Performing Health Provider Boards. *Mondaq*. Retrieved June 6 2020. <https://www.mondaq.com/canada/healthcare/839374/governance-best-practices-for-high-performing-health-provider-boards>.

Ditzel, E., P. Štrach and P. Pirozek. 2006. An Inquiry into Good Hospital Governance: A New Zealand-Czech Comparison. *Health Res Policy Sys* 4(2). doi: 10.1186/1478-4505-4-2.

Mays, N., J. Cumming and T. Tenbensel. 2007, August. *Report No. 1: Health Reforms 2001 Research: Overview Report*. Retrieved June 1, 2020. <https://www.wgtn.ac.nz/health/centres/health-services-research-centre/docs/reports/downloads/Report-No.-1-Overview-Report.pdf>.

National Research Council Canada (NRC). 2016. The Sentiment and Emotion Lexicons. Retrieved September 2019. <http://sentiment.nrc.ca/lexicons-for-research/>.

Pang B., and L. Lee. 2002. Thumbs Up? Sentiment Classification Using Machine Learning Techniques. Retrieved September 1, 2019. <https://www.cs.cornell.edu/home/llee/papers/sentiment.pdf>.

Peck, E. 1995. The Performance of an NHS Trust Board: Actors' Accounts, Minutes and Observation. *British Journal of Management* 6(2): 135–56. doi:10.1111/j.1467-8551.1995.tb00090.x.

Quigley, M.A. and G.W.S. Scott. 2004, April. *Hospital Governance and Accountability in Ontario*. Retrieved June 1, 2020. <https://www.longwoods.com/content/18348/law-and-governance/hospital-governance-and-accountability-in-ontario>.

Quigley, M. and G. Scott. 2005, September 13. *2nd Healthcare Governance Conference: Promoting and Enhancing Board Leadership. Roles and Responsibilities of the Board of Directors*. Retrieved June 1, 2020. https://www.longwoods.com/content/18353/law-and-governance/roles-responsibilities-healthcare-or-hospital-board-of-directors>.

Remy, P. (2020). Philipperemy/Name-Dataset. Retrieved January 2020. <https://github.com/philipperemy/name-dataset>.

Shen, M., S. Subramani, W. Zhao, J. Florian, J. Voqui, J. Meehan et al. 2018. A Semantic NLP Approach for Structuring and Analysis of FDA Meeting Minutes Documents. Retrieved Retrieved June 4, 2020. <https://www.lexjansen.com/phuse-us/2018/tt/TT07.pdf>.

About the Author

Lewis Hooper, BSc, is a healthcare executive with extensive experience in system planning. He has served on several boards both on the management side and the board side. Lewis also has extensive experience with technology and has worked on numerous large change projects. He can be reached by e-mail at Lewishooper@gmail.com.